

PU- Bootcamp Capstone Projects

What We Learned

Key Skills. Real Tools. End-to-End GenAI Solutions.



Programming & APIs

- Python
- OOP
- File handling
- FastAPI
- API validation
- Database integration



GenAI

- OpenAI APIs
- Prompt Engineering
- Advanced Prompting (CoT, ReAct, ToT)
- LangChain
- LangGraph
- LlamaIndex
- CrewAI
- AutoGen
- Function Calling
- Agents
- RAG
- Embeddings
- Vector Databases



Data Engineering

- ETL/ELT
- Data Cleaning
- Data Transformation
- SQL & NoSQL
- Data Ingestion



ML/MLOps

- ML Fundamentals
- Model Training
- Evaluation
- Deployment
- Docker
- MLOps



Cloud

- Azure Fundamentals
- Azure Storage
- Azure Databases
- Azure DevOps



AI-Powered GenAI Projects

1 AI-Powered Energy Consumption Optimization Assistant

Problem Statement

Energy consumers and facility managers struggle to understand consumption patterns and identify opportunities to reduce energy usage and costs.

Key Approach

- Ingest smart meter data & usage history
- RAG for energy efficiency best practices
- Consumption forecasting (ML)
- Personalized recommendations

Dataset (Example)

Building Energy Consumption Dataset
Format: CSV

2 AI-Powered Travel Planning & Itinerary Assistant

Problem Statement

Travelers spend significant time researching destinations, creating itineraries and finding reliable information across multiple sources.

Key Approach

- RAG for destination and travel information
- Hybrid search (keyword + semantic)
- Itinerary generation with LLMs/agents
- Personalized travel recommendations

Dataset (Example)

Kaggle Travel Dataset / TripAdvisor
Format: CSV / JSON

3 AI-Powered Agriculture Crop Advisory Assistant

Problem Statement

Farmers need timely and accurate advice on crop selection, disease management and best practices to improve yield and reduce losses.

Key Approach

- RAG for agricultural knowledge base
- Crop recommendation (ML)
- Disease detection guidance
- Personalized advisory and alerts

Dataset (Example)

Crop Recommendation Dataset
Format: CSV

4 AI-Powered Telecom Network Operations Assistant

Problem Statement

Network engineers take time to analyze network incidents, causing service disruptions and customer dissatisfaction.

Key Approach

- Ingest network logs & incident records
- RAG for troubleshooting guidance
- Incident classification & root cause analysis
- Resolution recommendations

Dataset (Example)

Telecom Network Incident Dataset
Format: CSV

5 AI-Powered Real Estate Property Valuation Assistant

Problem Statement

Buyers, sellers and investors find it challenging to estimate accurate property values and compare relevant properties.

Key Approach

- Ingest property data & market trends
- Hybrid search for comparable properties
- Price prediction (ML)
- Property recommendations

Dataset (Example)

Real Estate Property Dataset
Format: CSV

6 AI-Powered Restaurant Operations & Review Assistant

Problem Statement

Restaurant owners struggle to analyze customer reviews and operational data to improve service quality and efficiency.

Key Approach

- Ingest customer reviews & operational data
- Sentiment analysis (LLM/ML)
- RAG for best practices and menus
- Actionable insights and recommendations

Dataset (Example)

Restaurant Reviews Dataset
Format: CSV

7 AI-Powered Insurance Claims Assistant

Problem Statement

Claims processing is manual and time-consuming, leading to delays and customer dissatisfaction.

Key Approach

- Ingest claims data & policy documents
- Document understanding (LLM)
- Fraud/risk prediction (ML)
- Automated claim recommendations

Dataset (Example)

Insurance Claims Dataset
Format: CSV

8 AI-Powered Cybersecurity Incident Response Assistant

Problem Statement

Security analysts take time to analyze incidents, identify threats and respond effectively.

Key Approach

- Ingest security logs & threat intelligence
- Hybrid search for threat analysis
- Incident classification
- Response and mitigation recommendations

Dataset (Example)

CIC-IDS2017 / Security Logs
Format: CSV / Parquet

9 AI-Powered Supply Chain Disruption Assistant

Problem Statement

Supply chain disruptions impact delivery timelines, costs and customer satisfaction.

Key Approach

- Ingest supply chain & logistics data
- Disruption risk prediction (ML)
- RAG for mitigation strategies
- Alternative supplier recommendations

Dataset (Example)

Supply Chain Disruption Dataset
Format: CSV

10 AI-Powered Education Student Success Assistant

Problem Statement

Students at risk of poor performance are not identified early, leading to higher dropout rates.

Key Approach

- Ingest student academic & activity data
- Performance prediction (ML)
- RAG for personalized learning guidance
- Early intervention recommendations

Dataset (Example)

OULAD (Open University Learning Analytics)
Format: CSV

COMMON TECHNIQUES ACROSS ALL PROJECTS



Data Ingestion
& Preprocessing



Text Chunking
& Parsing



Embeddings
Generation



Vector Database
(ChromaDB / FAISS)



Hybrid Search
(Keyword + Semantic)



LLM / RAG
(Prompt Engineering)



Guardrails &
Validation



API
(REST)



Evaluation &
Continuous Learning



Build Smart Solutions. Solve Real Problems. Create Real Impact.

Learn | Implement | Innovate | Make a Difference

Sample Project

A GenAI solution for smarter marketing decisions

“Turn campaign data into actionable insights with GenAI”



AI-Powered Marketing Campaign Optimization Assistant

An end-to-end GenAI solution for campaign analysis, customer targeting and performance optimization



Project Overview

Marketing teams spend significant time analyzing previous campaigns, customer segments, contact history and campaign outcomes to determine which strategies are likely to perform well. Historical campaign data can be difficult to search and compare manually.



Marketing teams often ask:

- Which previous campaigns had similar customer profiles?
- What factors contributed to campaign success or failure?
- Which customers or segments should be prioritized for the next campaign?
- What campaign strategy is likely to improve conversion?



Requirement 1 – Basic

- ✓ Basic RAG for campaign similarity search
- ✓ Hybrid search (keyword + semantic)
- ✓ Campaign/customer response classification agent
- ✓ Metadata filtering (segment, job, education, contact type, month, previous outcome)
- ✓ Campaign performance recommendation
- ✓ Input validation guardrails
- ✓ Campaign/customer prioritization
- ✓ Expose functionality through API endpoints



Requirement 2 – Advanced

- ✓ DeepEval for recommendation quality evaluation
- ✓ Campaign response/conversion prediction
- ✓ LLM-as-Judge validation
- ✓ Multi-agent campaign optimization workflow
- ✓ Customer response and campaign-performance analysis
- ✓ Explainable analysis of factors influencing campaign success
- ✓ Continuous feedback learning
- ✓ Simple front-end interface (Streamlit)



Solution Architecture (High Level)



Data Ingestion

- Campaign data
- Customer data
- Preprocessing



Embeddings & Vector Store

- Text + structured data
- Vector database (ChromaDB / FAISS)



RAG Pipeline

- Hybrid search
- Context retrieval
- Similarity search



AI Agents

- Campaign analysis
- Recommendation
- Multi-agent workflow



ML / Prediction

- Response prediction
- Customer ranking
- Explainable AI



API Layer

- FastAPI
- REST endpoints
- Swagger docs



Frontend (UI)

- Streamlit
- Search & explore
- Insights & recommendations



Dataset Details

Dataset: UCI Bank Marketing Dataset

Format: CSV

Key Fields:

age, job, marital, education, balance, housing, loan, contact, month, campaign, pdays, previous, poutcome, y

Records: 45,211 (UCI Repository)



Sample User Query & Output

User Query:

“Find previous campaigns with similar customer profiles and recommend an approach for customers with high probability of subscribing.”



System Output:

- Similar campaign/customer cases
- Predicted response probability
- Key factors influencing success
- Recommended campaign approach
- Supporting historical evidence



Business Impact

- ✓ Higher campaign conversion rates
- ✓ Better customer targeting
- ✓ Reduced marketing costs
- ✓ Data-driven campaign strategy
- ✓ Faster access to campaign insights
- ✓ Improved ROI through AI recommendations



**Smarter campaigns.
Higher conversions. Greater ROI.**



AI-Powered Telecom Network Fault Resolution Assistant

Milestone Plan (7-Day Project Plan in 2 Milestones)

An end-to-end GenAI solution for intelligent fault search, root cause analysis and resolution



1

MILESTONE 1 – Core Fault Intelligence System



Oct 5 – 7



OBJECTIVE: Build a working end-to-end solution that can retrieve similar telecom incidents, classify severity, identify probable root causes and suggest initial resolutions.



DATA LAYER

- Network logs ingestion
- Incident data cleaning & preprocessing
- Metadata extraction
- Log normalization



KNOWLEDGE LAYER

- Text + structured data processing
- Embeddings generation
- Vector database setup (Pinecone/ ChromaDB/FAISS)
- Incident indexing



RAG LAYER

- Similarity search
- Hybrid search (keyword + semantic)
- Context retrieval
- Relevant past incidents, KB articles and manuals



INTELLIGENCE LAYER

- Incident classification (e.g., severity, type)
- Probable root cause identification
- Initial resolution recommendations
- Input validation safeguards



API LAYER

- FastAPI endpoints
- Input validation
- Swagger docs
- Unit testing
- Basic UI (Streamlit) for query interface

DELIVERABLES (Milestone 1)



Working RAG system



Fault similarity search



Root cause recommendations



Resolution suggestions



API endpoints



DEMO OUTCOME



Engineer enters:

"Customers are facing frequent call drops in Chennai"



System returns:

- Similar incidents
- Root cause
- Resolution steps

2

MILESTONE 2 – Advanced AI, Evaluation & Final Deliverables



Oct 8 – 11



OBJECTIVE: Enhance the core solution with agentic workflows, deep evaluation, monitoring, and a user-friendly interface, and prepare final project deliverables.



AGENTIC AI

- Multi-agent workflow (L1 → L2 → L3)
- Agent communication
- Escalation routing



ADVANCED AI FEATURES

- Resolution time prediction
- Impact analysis
- Proactive alerts
- Explainable AI (key factors)



EVALUATION

- DeepEval integration
- LLM-as-Judge validation
- Retrieval quality metrics (relevance, accuracy)
- Feedback loop



PRODUCTION READINESS

- Logging & monitoring
- Error handling
- Fault alert dashboard
- Performance testing
- Security considerations
- Deployment (optional)



FINAL DELIVERABLES

- Complete source code
- Architecture diagram
- Design document
- Evaluation report
- README
- Presentation deck
- Demo video (optional)

DELIVERABLES (Milestone 2)



Multi-agent system



DeepEval integration



Monitoring dashboard (Logging & alerts)



Architecture diagram



Final presentation



BUSINESS IMPACT

- ✓ Faster fault resolution and reduced downtime
- ✓ Improved network reliability and customer satisfaction
- ✓ Reduced manual troubleshooting effort
- ✓ Better knowledge reuse and consistency
- ✓ Data-driven insights for proactive maintenance
- ✓ Scalable solution for real-time incident support



Smarter networks.
Faster resolutions.
Higher availability.
Greater customer satisfaction.

MILESTONE SUMMARY

1

Milestone 1

Core RAG-based Telecom Fault Resolution System



~70%
Project

2

Milestone 2

Agentic AI, Evaluation, Monitoring & Final Deliverables



~30%
Project



Together, these two milestones deliver the complete AI-powered solution as defined in the project requirements.

Milestones



ADD-ON TECHNICAL SKILLS (RECOMMENDED)



These 3 add-on skills will help you build a more accurate, reliable and production-ready **AI-Powered Telecom Network Fault Resolution Assistant**.

ADD-ON SKILL

WHY IT HELPS

WHAT YOU WILL LEARN



1. Vector Databases & Advanced RAG

Improve fault similarity search and retrieval accuracy.



Enables the system to find the most relevant historical faults and solutions using semantic understanding and hybrid search, leading to better root cause identification and resolutions.



- Embeddings
- Semantic Search
- Hybrid Search (Keyword + Semantic)
- Reranking
- Metadata Filtering



2. LLM Evaluation & Guardrails

Measures response quality and reduces hallucinations.



Helps evaluate the quality, relevance and correctness of LLM responses and retrieved context. Ensures safe, reliable and trustworthy outputs for network engineers.



- DeepEval for RAG Evaluation
- Ragas Metrics
- Groundedness & Context Relevance
- Answer Relevance
- LLM-as-Judge Validation



3. Testing, Monitoring & Observability

Build reliable, testable and production-ready GenAI applications.



Ensures system reliability, easy debugging and continuous performance improvement through monitoring, logging and testing.



- API Testing & Validation
- Logging & Error Tracking
- Performance Monitoring
- Alerts & Observability Dashboards



OUTCOME: These add-on skills will significantly improve retrieval accuracy, response quality, system reliability and overall solution impact.



TIP: Focus on these skills in parallel with the project implementation for maximum value.

Coursera Courses

Add-On Skill	Why It Helps	Course Name	Instructor	Provider
Vector Databases & Advanced RAG	Improves retrieval accuracy, semantic search, hybrid search, and RAG performance	Generative AI Engineering with LLMs Specialization	Multiple Instructors	IBM
LLM Evaluation & Guardrails	Learn evaluation metrics, hallucination detection, and response quality assessment	Generative AI with Large Language Models	Sharon Zhou, Mike Loukides	DeepLearning.AI & AWS
Testing, Monitoring & Observability	Build reliable, testable, production-ready AI applications	Software Testing and Automation Specialization	Amir Ghahrai	University of Minnesota

GenAI Capstone Project – Evaluation Rubric (Total: 100 Marks)

#	Evaluation Criteria		What We Look For	How We Evaluate	Weightage (%)
1		Functional Correctness & Deliverables	- Completeness of solution - All required features working end-to-end	- Validate workflow, stability, reliability - Check output accuracy	20%
2		Architecture & Design Decisions	- Architecture quality, modularity - Maintainability, scalability	Review architecture and design justifications	10%
3		LLM Selection	- Model choice and rationale - Cost vs capability	Evaluate suitability and trade-offs	5%
4		Embedding Model Evaluation	- Embedding selection - Impact on retrieval quality	Measure retrieval accuracy and efficiency	5%
5		Retrieval & RAG Strategy	- Chunking, retrieval, Top-K - Context construction and grounding	Evaluate retrieval quality and relevance	10%
6		Accuracy & Quality Evaluation	- Groundedness, completeness - Hallucination rate and response quality	Use RAGAS/DeepEval and human evaluation	10%
7		Agent Design & Orchestration	- Agent workflow, tools - Multi-agent collaboration and reasoning	Assess reasoning, planning and tool usage	10%
8		Security & Compliance	- Input validation, privacy - Prompt injection protection and compliance	Review security and compliance measures	5%
9		User Experience	- Readability, consistency - Transparency and ease of use	Evaluate usability and satisfaction	5%
10		Innovation & Optimization	- Novel ideas and enhancements - Efficiency improvements and unique features	Assess uniqueness and impact	5%
11		Presentation & Documentation	- README, technical documentation - Demo quality and communication	Review presentation and documentation quality	10%
12		Testing & Robustness	- Test coverage, edge cases - Error handling and reliability	Validate error handling and robustness	5%
Total					100%



PARTICIPANT EXPECTATIONS & SUBMISSION GUIDELINES



AI-Powered Telecom Network Fault Resolution Assistant – Sample Project

EXPECTATIONS FROM PARTICIPANTS



Understand the Problem

Thoroughly understand the telecom use cases and build solutions that address real-world challenges.



Innovate & Add Value

Go beyond the basics. Use GenAI, agents, and data to create intelligent, impactful solutions.



Build High-Quality Solutions

Ensure your solution is accurate, reliable, scalable, and user-friendly.



Write Clean & Modular Code

Follow best coding practices. Code should be well-structured, readable, and easy to run.



Document Everything

Provide clear documentation for setup, usage, APIs, assumptions, and design decisions.



Test & Validate

Evaluate your solution using appropriate metrics/tools (DeepEval, test cases, etc.).



Work Ethically & Collaboratively

Original work only. Acknowledge references. Collaborate within your team effectively.



Present with Clarity & Confidence

Prepare a concise presentation to explain your solution, design, and impact.

SUBMISSION PROCESS



Step 1

Prepare Files

Organize all required files as per the submission checklist.

- Source Code
- Documents
- Diagrams
- Datasets
- Presentation
- Others



Step 2

Compress as ZIP

Compress the complete project folder into a single ZIP file.



Step 3

Upload to GitHub

Upload the ZIP file in your existing GitHub repository.



Step 4

Confirmation

Ensure all files are uploaded successfully and accessible to the evaluators.



Submit:

Share your GitHub Repository URL with the organizers along with the following:

- GitHub Repository URL
- Demo Video URL (Optional)
- Team Details (Names, ID, Email)

REQUIRED SUBMISSION FILES & CONTENT



1. Architecture Diagram (PDF/JPEG)

- High-level system architecture
- Show data flow: ingestion, RAG pipeline, agents, APIs, and UI

File Name: 01_Architecture.pdf



2. Design Document (PDF)

- Design decisions
- Technology choices
- Trade-offs
- Scalability considerations

File Name: 02_Design_Document.pdf



3. Source Code (Folder/ZIP)

- Complete source code of the project
- Well-structured and modular
- Include all dependencies

Folder Name: 03_Source_Code/



4. README File (Markdown/PDF)

- Project overview
- Setup instructions
- How to run
- API endpoints
- Sample queries & outputs

File Name: README.md



5. API Documentation (PDF/Markdown)

- List of all APIs
- Request format
- Response format
- Example requests & responses

File Name: 05_API_Docs.pdf



6. Dataset Details (PDF/CSV)

- Dataset name & description
- Source (if any)
- Key fields description
- Sample data

File Name: 06_Dataset_Details.pdf



7. Demonstration Video (MP4)

- Working demo of the solution
- Max 5-7 minutes
- Upload link (Google Drive / YouTube Unlisted)

File Name: 07_Demo_Video.mp4
(Or provide sharing link)



8. Presentation Deck (MP4/PDF)

- 10-minute presentation
- Problem, approach, demo, results, impact

File Name: 08_Presentation.pdf

RECOMMENDED FOLDER STRUCTURE (INSIDE SOURCE CODE FOLDER)



/app
Main application code



/agents
Agent implementations



/data
Data ingestion scripts



/docs
Documentation files



/tests
Test cases & examples



/utils
Utility functions



requirements.txt
Dependencies list

SUBMISSION CHECKLIST (BEFORE UPLOADING)

- ✓ All required files are included.
- ✓ Code runs successfully as per README.
- ✓ APIs are tested and working.
- ✓ Architecture & design are clear and complete.
- ✓ Demo video and presentation are included.
- ✓ Files are zipped and named correctly.



Build Smart Solutions. Solve Real Problems. Create Real Impact.

We can't wait to see what you build!



Q & A



Thank you

